Intracellular Accumulations

SGL – 2

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Objectives:

1. To study the types and mechanism of intracellular accumulation.

2. To explain causes and morphology of fatty changes of liver.

3. To define hemosiderosis and pathological calcification.

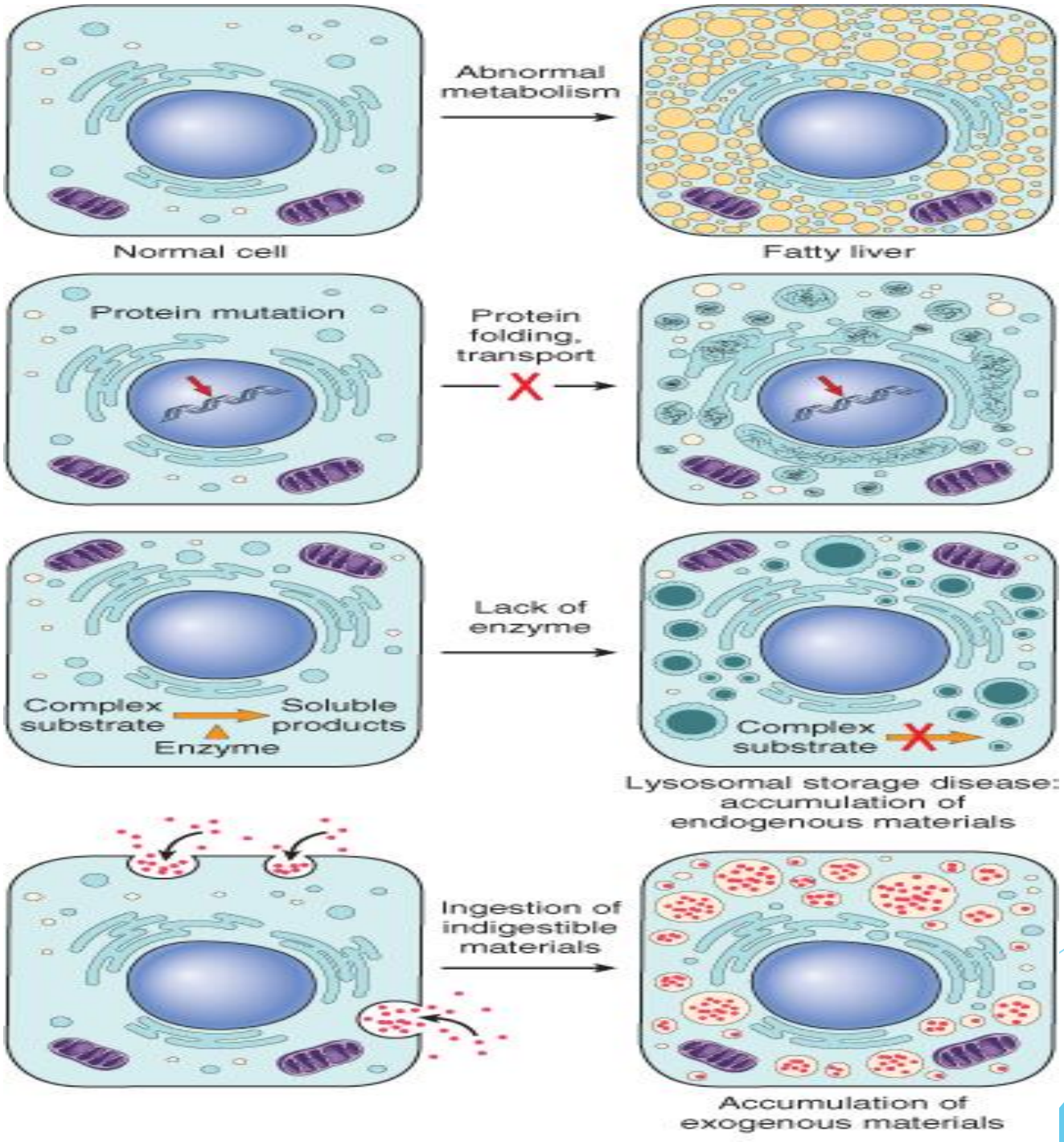
- Under some circumstances cells may accumulate abnormal amounts of various substances, which may be harmless or associated with varying degrees of injury.
- The substance may be located in the cytoplasm, within organelles (typically lysosomes), or in the nucleus, and it may be synthesized by the affected cells or may be produced elsewhere.

Many processes result in abnormal intracellular accumulation, but most accumulations are attributable to three types of abnormalities:

1. A normal endogenous substance (water, protein, carbohydrate, lipid) is produced at a normal (or even increased) rate but the metabolic rate is inadequate to remove it (e.g. fat accumulation in liver cells).

2. A normal or abnormal endogenous substance accumulates because of genetic or acquired defects in metabolism, Packaging, transport, or secretion of these substances (e.g. lysosomal storage diseases, or α1 -antitrypsin disease).

3. Abnormal exogenous substances may accumulate in normal cells because they lack the machinery to degrade such substances (e.g. macrophages laden with environmental carbon).


Abnormal metabolism
Normal cell
Fatty liver
Protein mutation
Protein folding, transport
Lack of enzyme
Complex substrate
Soluble products
Enzyme
Complex substrate
Lysosomal storage disease: accumulation of endogenous materials
Ingestion of indigestible materials
Accumulation of exogenous materials

Fatty Change (Steatosis):

- Fatty change refers to any abnormal accumulation of triglycerides within parenchymal cells.
- It is most often seen in the liver, since this is the major organ involved in fat metabolism, but it may also occur in heart, skeletal muscle, kidney, and other organs.

- may be caused by Alcohol, protein malnutrition, diabetes mellitus, obesity, and anoxia.
- The significance of fatty change depends on the cause and severity of the accumulation.
- When mild it may have no effect on cellular function.
- More severe fatty change may transiently impair cellular function.

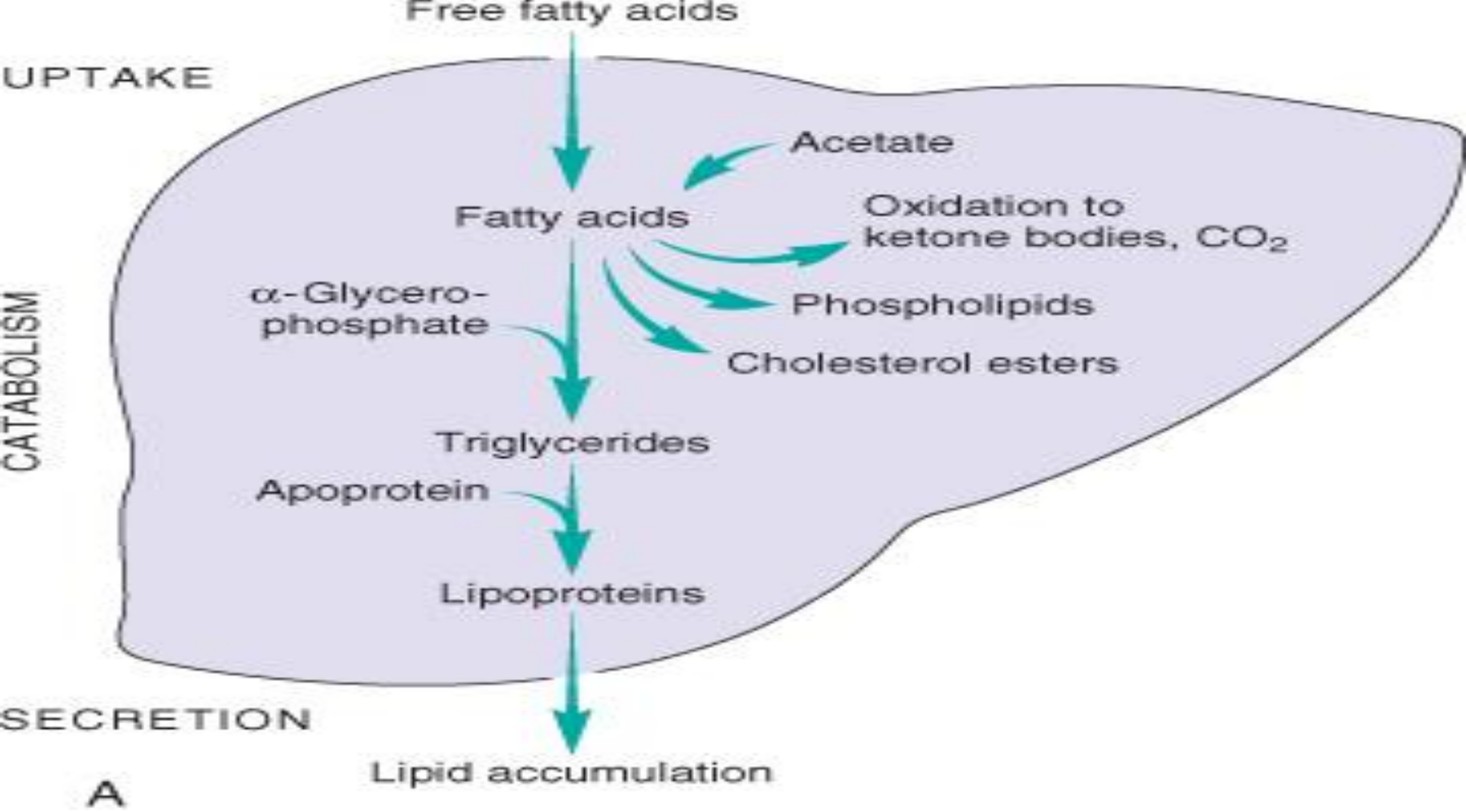


Diagram of the possible mechanisms leading to accumulation of triglycerides in fatty liver.

Any defects in any of the steps of uptake, catabolism or secretion can result in lipid accumulation.

Morphology

- In any site, fatty accumulation appears as clear vacuoles within parenchymal cells.
- Special staining techniques are required to distinguish fat from intracellular water or glycogen, which can also produce clear vacuoles.
- To identify fat microscopically, tissues must be processed for sectioning without the organic solvents typically used in sample preparation.

- Usually, portions of tissue are therefore frozen to enable the cutting of thin sections for histologic examination; the fat is then identified by staining with Sudan IV or oil red O (these stain fat orange-red).
- Fatty change is most commonly seen in the liver and the heart.

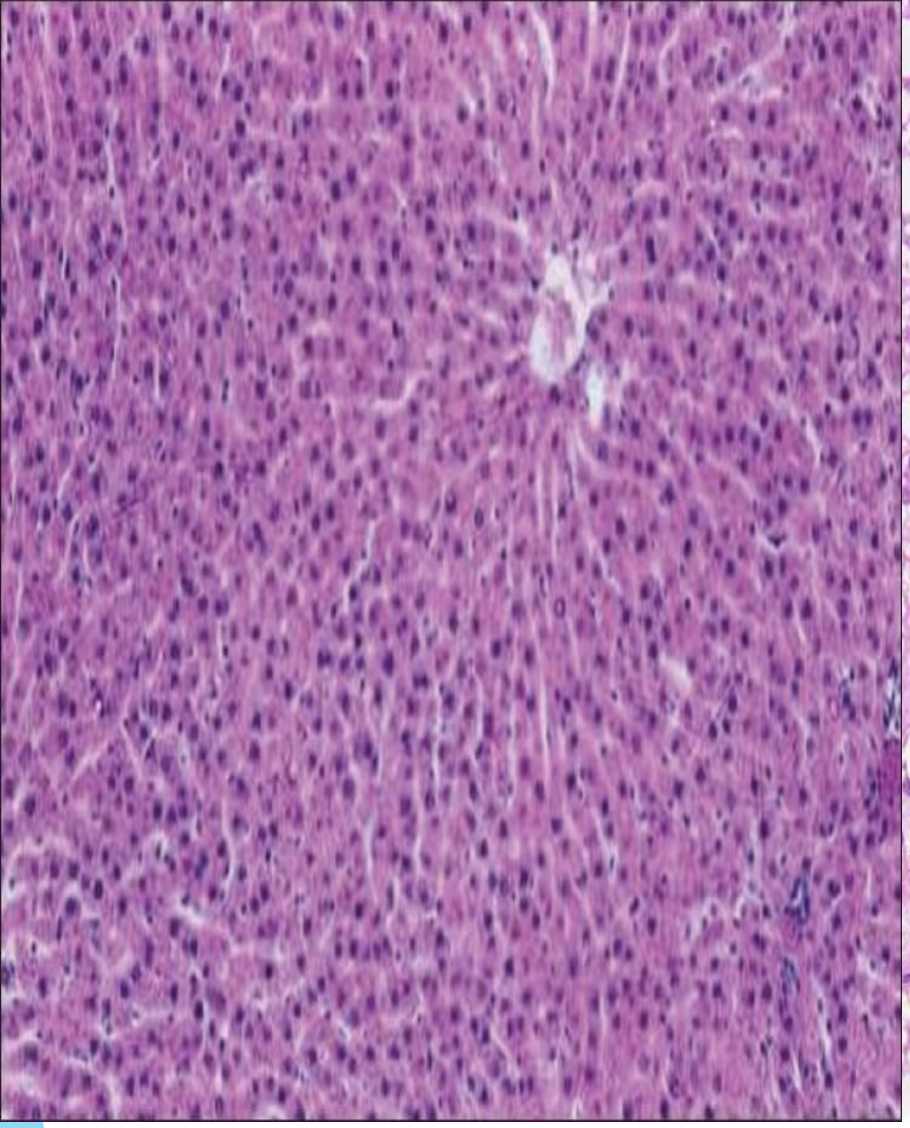
- **gross appearance: the organ enlarges and becomes progressively yellow until, in extreme cases, it may weigh 3 to 6 kg (1.5-3 times the normal weight) and appear bright yellow, soft, and greasy.**
- **light microscopy: Early fatty change is seen as small fat vacuoles in the cytoplasm around the nucleus. In later stages, the vacuoles coalesce to create cleared spaces that displace the nucleus to the cell periphery.**



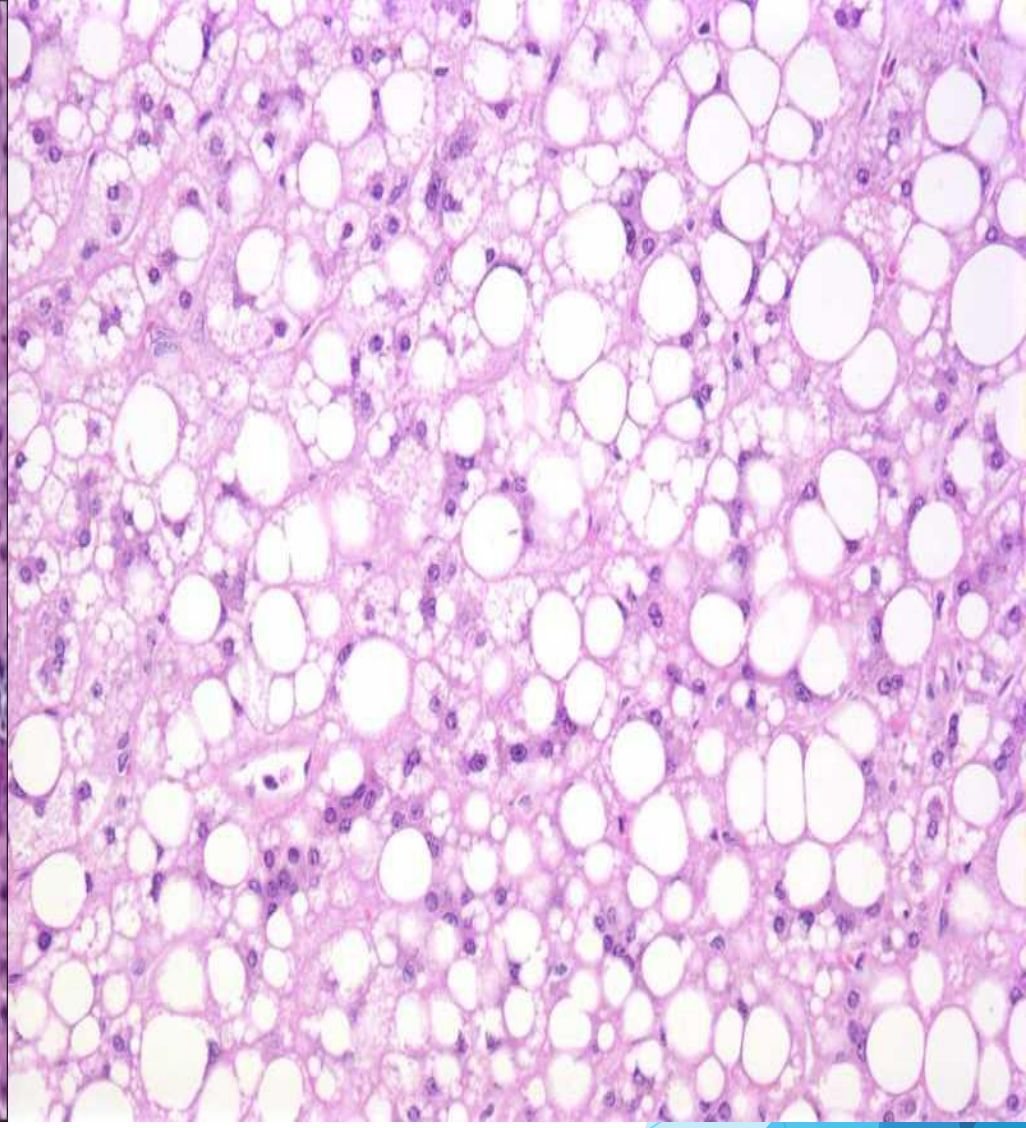
Liver,
normal

Liver, fatty
change





Normal liver histology



Microscopical appearance of fatty change of liver

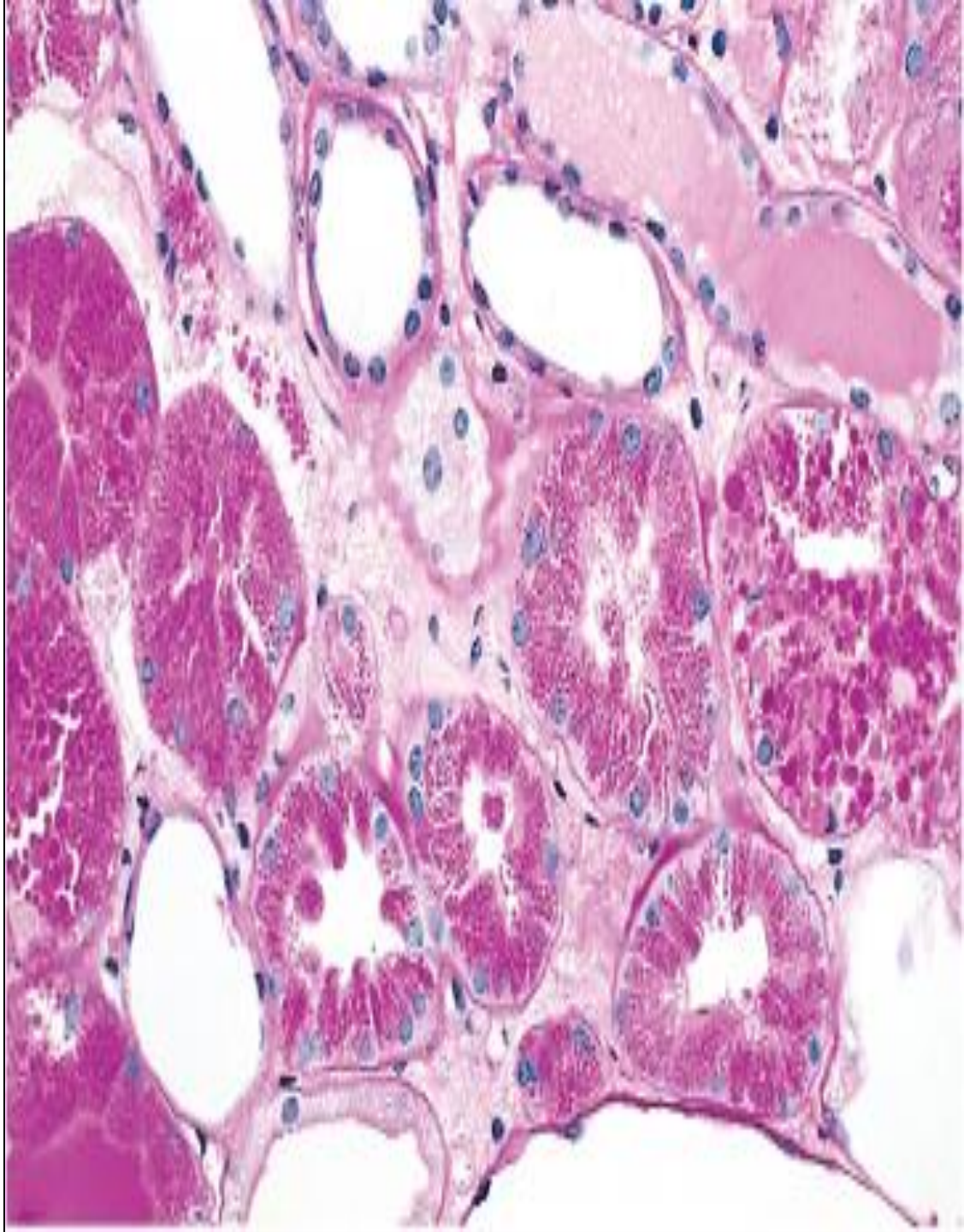
Proteins:

Intracellular protein accumulation may be due to excessive synthesis, absorption, or defects in cellular transport.

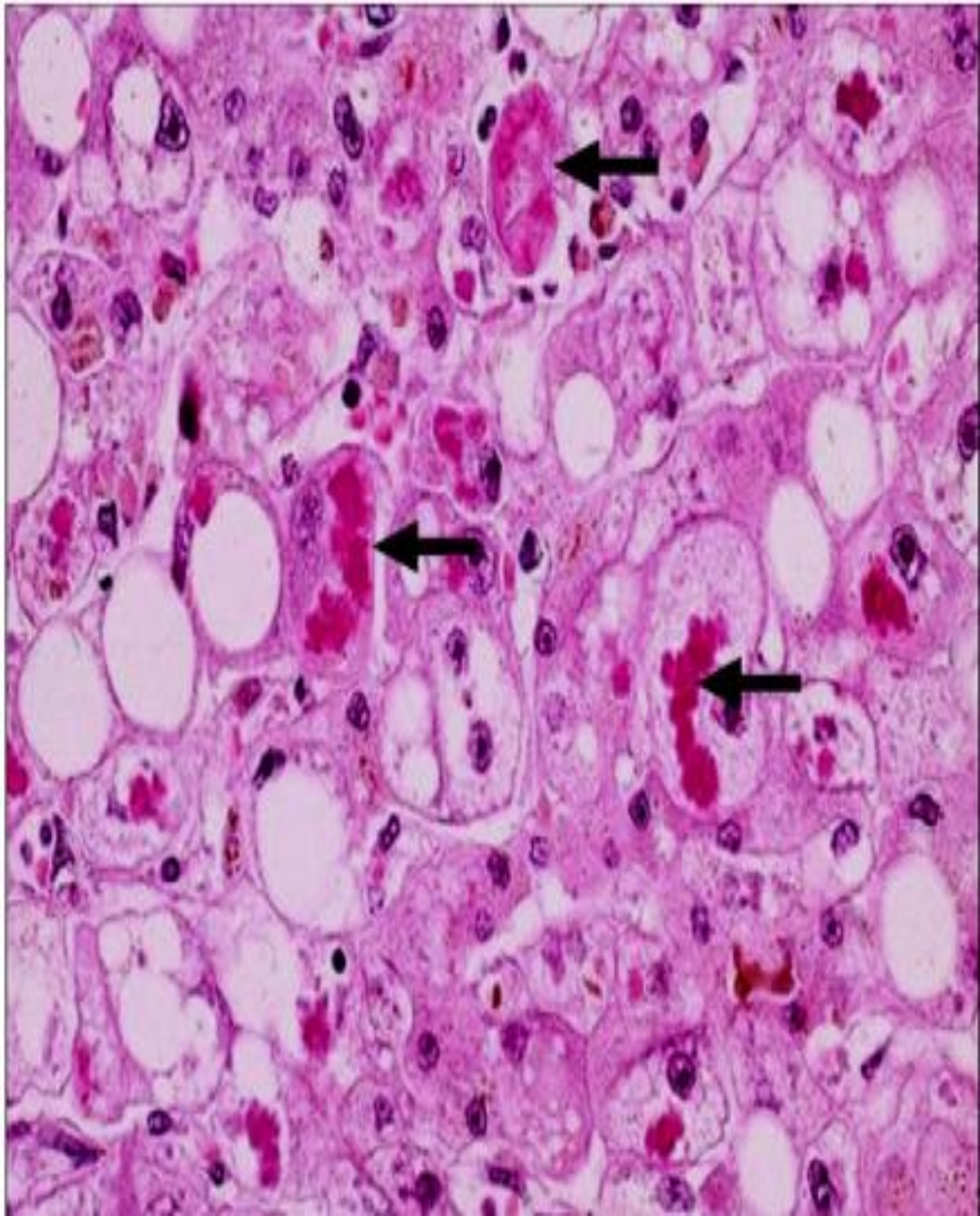
Morphologically visible accumulation appear as rounded, eosinophilic, cytoplasmic droplets.

E.g. : protein in proximal convoluted tubular epithelium due to chronic reabsorption in the setting of proteinuria.

In **proteinuria**, there is excessive **renal tubular reabsorption** of proteins by the proximal tubular epithelial cells which show **pink hyaline droplets** in their cytoplasm.



Mallory's body or **alcoholic hyalin** in the hepatocytes is intracellular accumulation of **intermediate filaments** of **cytokeratin**.



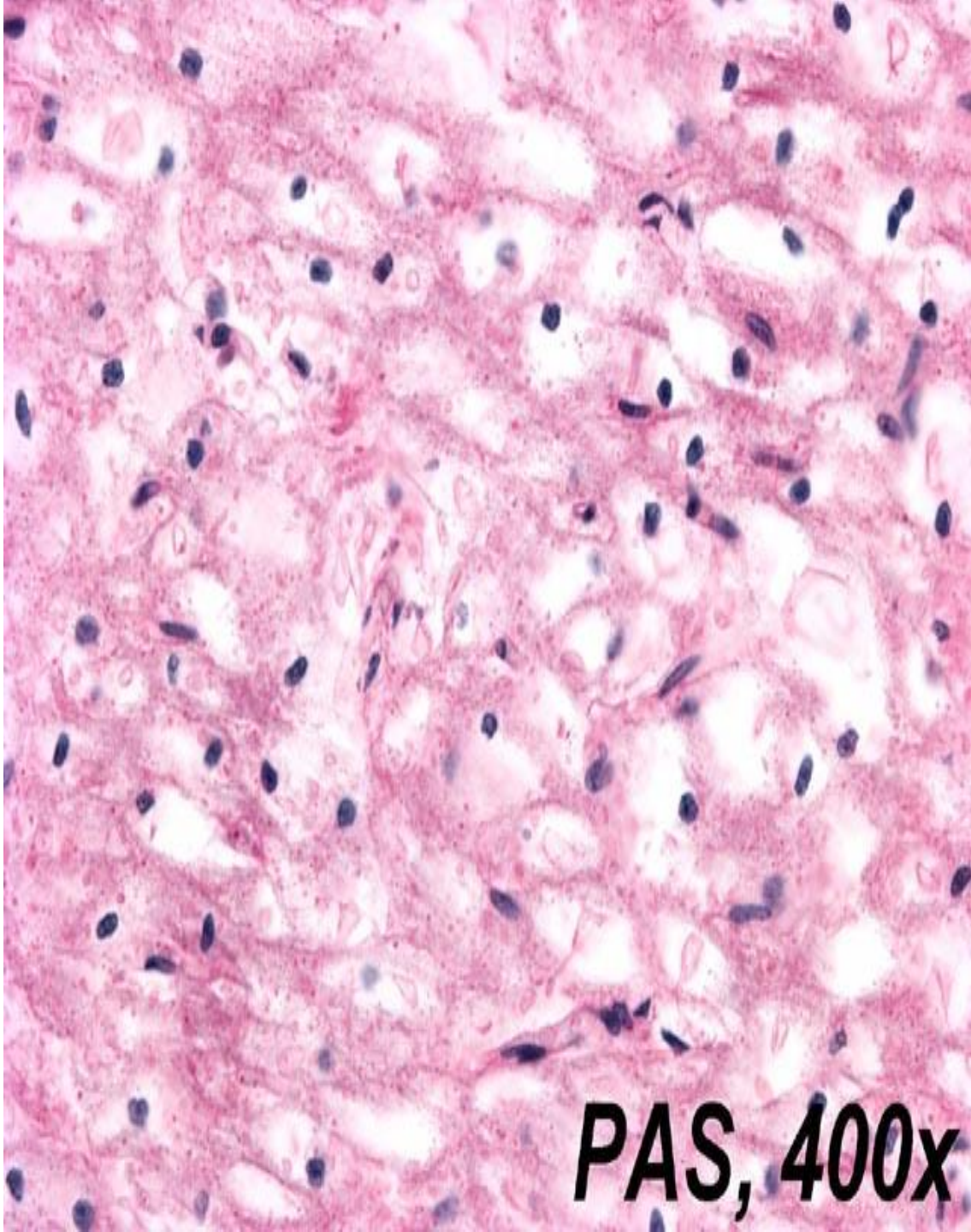
Glycogen:

Is a readily available cytoplasmic energy store.

Excessive intracellular deposits which seen as clear vacuoles, are seen with abnormalities of glycogen storage so called glycogenosis, and glucose metabolism as in diabetes mellitus.

Glycogen storage disease (Glycogenosis)

- genetic diseases
- defect of enzymes in the synthesis or breakdown glycogen

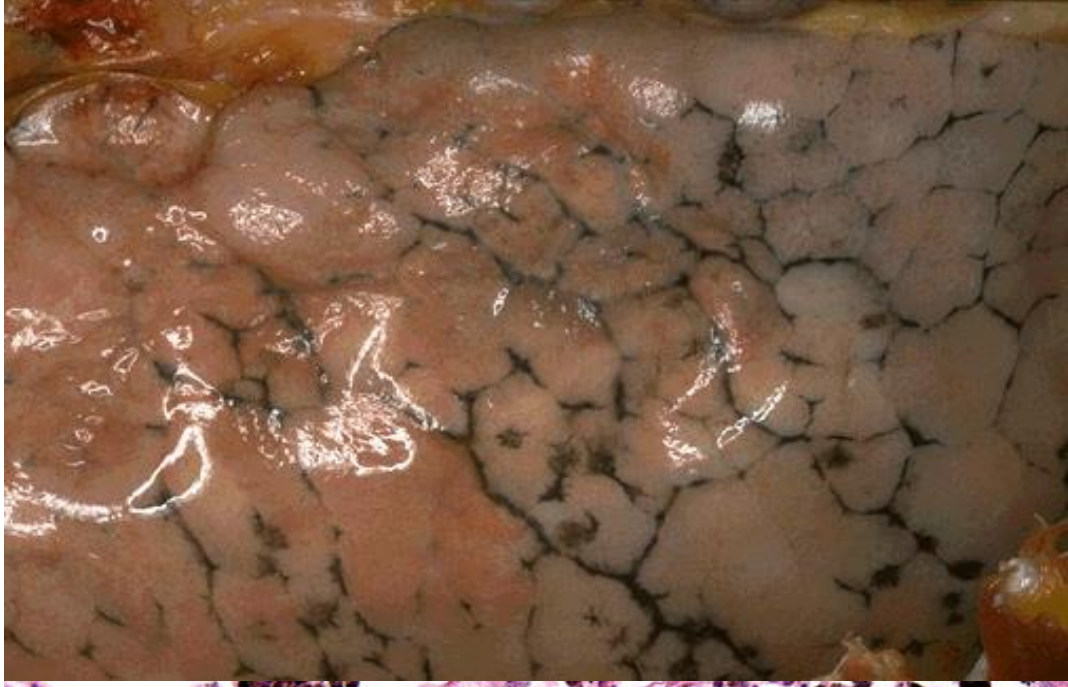


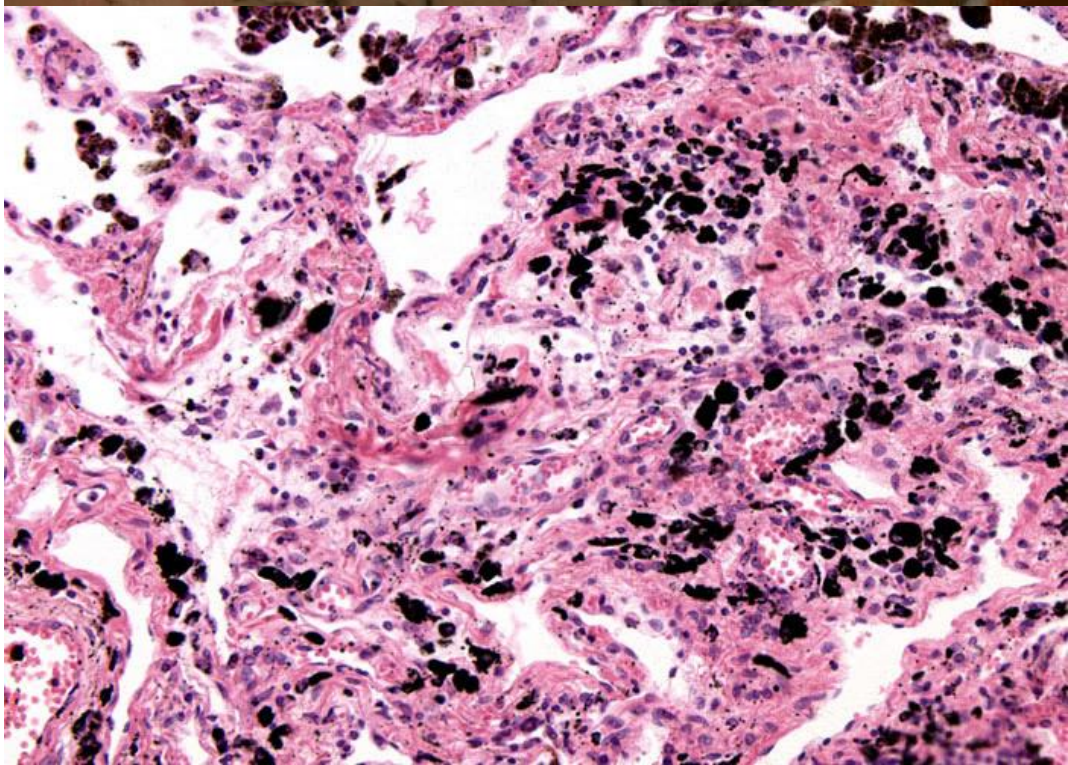
Pigments:

- Pigments are colored substances that are either exogenous, coming from outside the body, or endogenous, synthesized within the body itself.
- The most common exogenous pigment is carbon (an example is coal dust), a common air pollutant of urban life.
- When inhaled, it is phagocytosed by alveolar macrophages and transported through lymphatic channels to the regional tracheobronchial lymph nodes.

- Aggregates of the pigment blacken the draining lymph nodes and pulmonary parenchyma (anthracosis).
- Heavy accumulations may induce emphysema or a fibroblastic reaction that can result in a serious lung disease called coal workers' pneumoconiosis.







Anthracosis of Lung

Endogenous pigments

Include:

- **Lipofuscin**
- **Melanin**
- **Hemosiderin.**

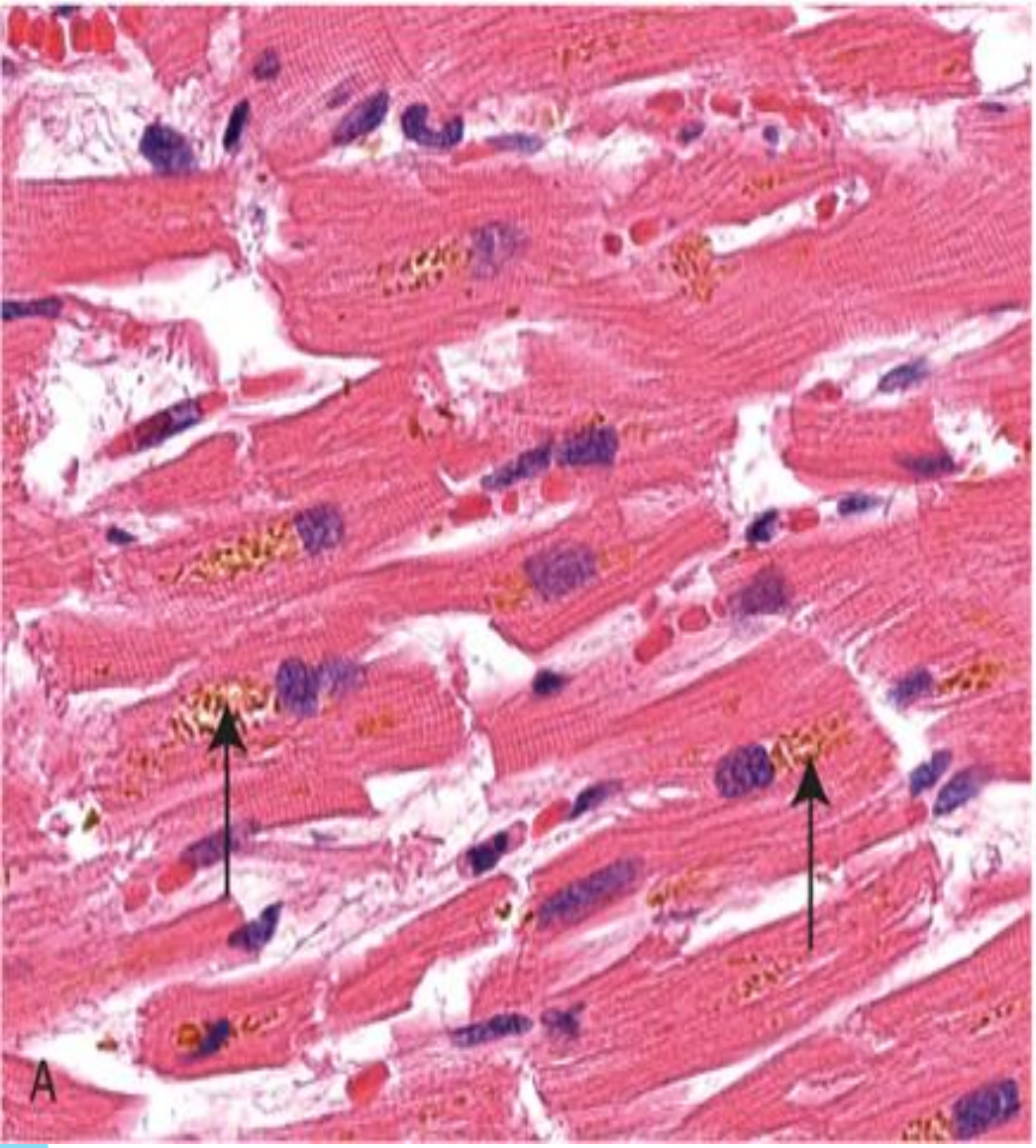
- **Lipofuscin:**

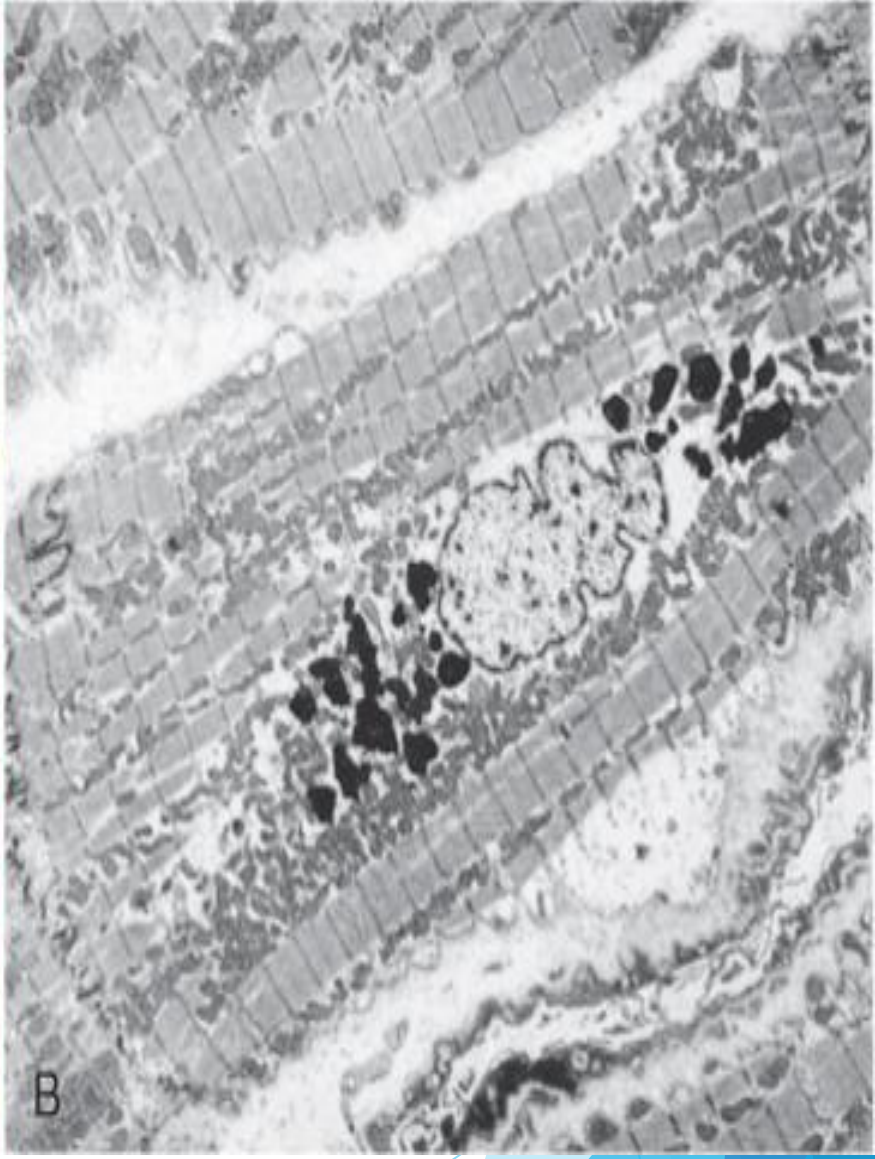
is an insoluble brownish-yellow granular intracellular material that accumulates in a variety of tissues (particularly the heart, liver, and brain) as a function of age or atrophy.

- Lipofuscin represents complexes of lipid and protein that derive from the free radical- injury.

- The brown pigment when present in large amounts, imparts an appearance to the tissue that is called brown atrophy.

- By electron microscopy, the pigment appears as perinuclear electron-dense granules.





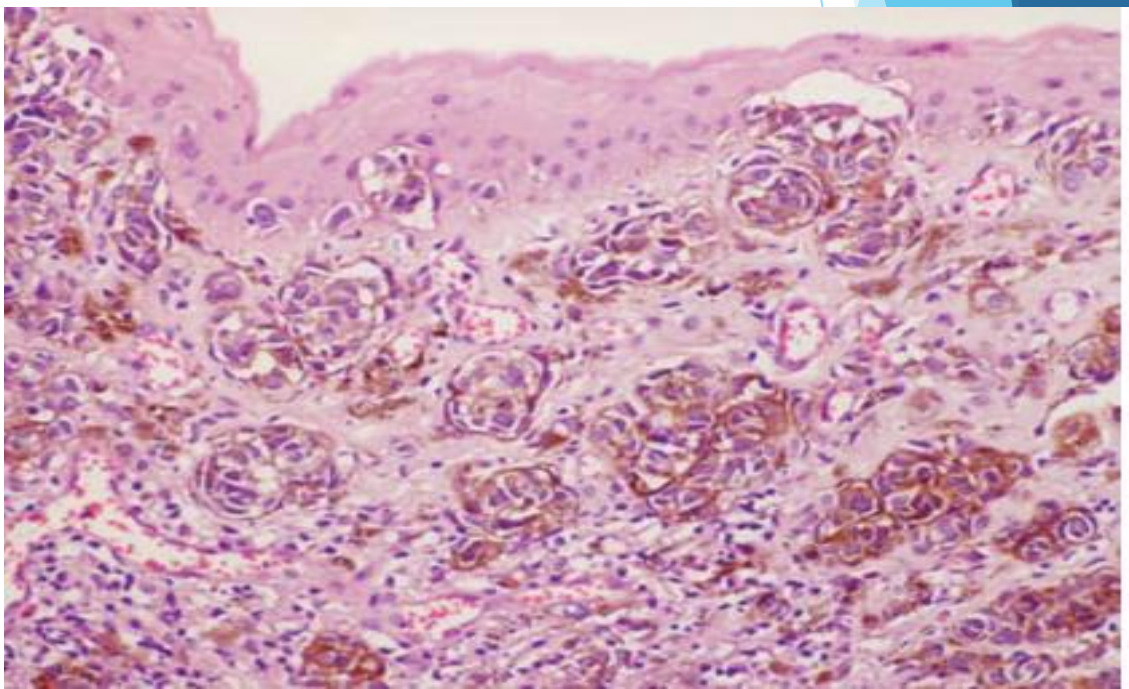
Lipofuscin granules in cardiac myocytes

Electron microscopy
note prinuclear intralysosomal location

Melanin :

- is an endogenous, brown-black pigment.

It is synthesized exclusively by melanocytes located in the epidermis and acts as a screen against harmful ultraviolet radiation.



Hemosiderin:

- is a hemoglobin-derived granular pigment that is golden yellow to brown and accumulates in tissues when there is a local or systemic excess of iron.
- Iron is normally stored within cells in association with the protein *apoferritin*, forming ferritin.

Hemosiderin pigment represents large aggregates of ferritin, readily visualized by light and electron microscopy; the iron can be identified by the Prussian blue histochemical reaction.

- Although hemosiderin accumulation is usually pathologic, small amounts of this pigment are normal in the mononuclear phagocytes of the bone marrow, spleen, and liver, where there is extensive red cell breakdown.
- Local excesses of iron, and consequently of hemosiderin, result from hemorrhage.

The best example is the common bruise.

Hemosiderosis:

- A condition when there is systemic overload of iron, hemosiderin is deposited in many organs and tissues.
- It is found at first in the mononuclear phagocytes of the liver, bone marrow, spleen, and lymph nodes and in scattered macrophages throughout other organs.

With progressive accumulation, parenchymal cells throughout the body (but principally the liver, pancreas, heart, and endocrine organs) become "bronzed" with accumulating pigment.

Hemosiderosis occur in the setting of:

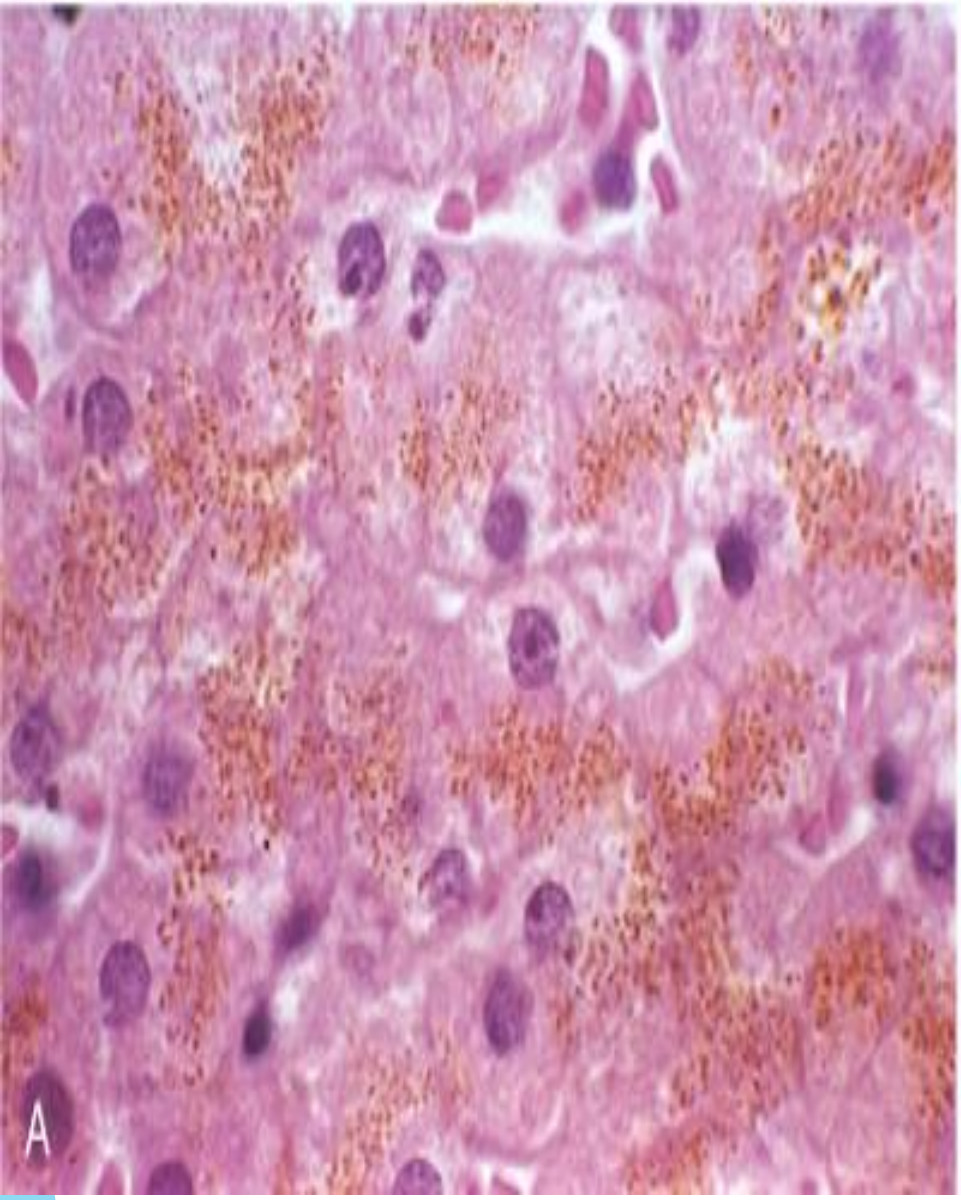
(1) increased absorption of dietary iron,

(2) impaired utilization of iron,

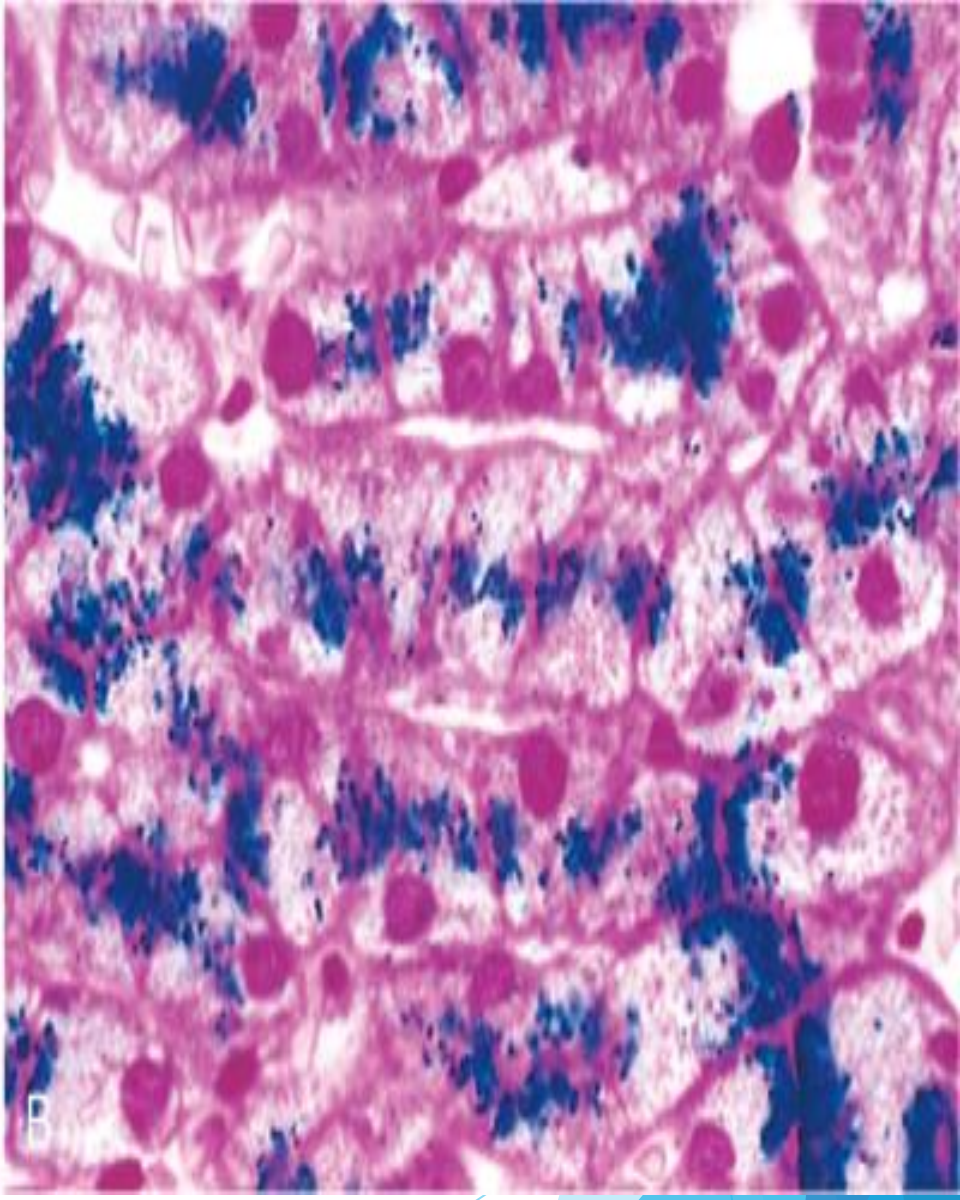
(3) hemolytic anemias.

(4) transfusions (the transfused red cells constitute an exogenous load of iron).

In most instances of systemic hemosiderosis, the iron pigment does not damage the parenchymal cells or impair organ function despite an impressive accumulation.



Hemosiderin granules in hepatocytes



Prussian blue histochemical reaction

PATHOLOGIC CALCIFICATION

- Pathologic calcification is a common process in a wide variety of disease states;
- it implies the abnormal deposition of calcium salts, together with smaller amounts of iron, magnesium, and other minerals.

- When the deposition occurs in dead or dying tissues, it is called dystrophic calcification; it occurs in the absence of calcium metabolic derangements (i.e., with normal serum levels of calcium).
- In contrast, the deposition of calcium salts in normal tissues is known as metastatic calcification and almost always reflects some derangement in calcium metabolism (hypercalcemia).

Dystrophic calcification:

- Dystrophic calcification is encountered in areas of necrosis of any type.
- It is virtually inevitable in the atheromas of advanced atherosclerosis, associated with intimal injury in the aorta and large arteries and characterized by accumulation of lipids.

Dystrophic calcification of the aortic valves is an important cause of aortic stenosis in the elderly.



Metastatic Calcification:

- Metastatic calcification can occur in normal tissues whenever there is hypercalcemia.
- The four major causes of hypercalcemia are

1. increased secretion of parathyroid hormone,
2. destruction of bone due to the effects of accelerated turnover
3. vitamin D-related disorders including vitamin D intoxication
4. renal failure, in which phosphate retention leads to secondary hyperparathyroidism.

Thank You